

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a vast, blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene, digital aesthetic.

DL for ENG

LECTURE 9.2

Gas Flow

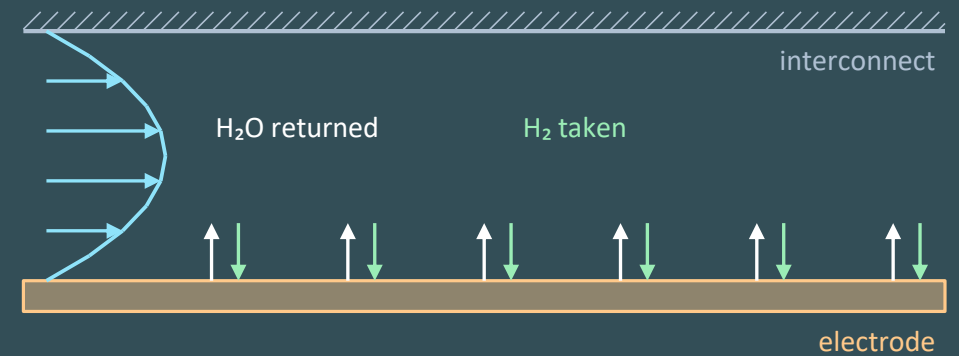
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Deep Learning for Engineering · 2026

GAS · SPECIES · EVERY SPEED



The fuel channel of a solid oxide cell. Three facts make its gas behave like the liquid of the last lecture, so the flow is known; the question is what the gas carries - hydrogen in, steam out - and one PINN, with the speed as an input, answers it for every speed and finds the speed a design asks for.

Learning Objectives

SECTIONS, IN ORDER

- 1 **The gas** — slow, one for one, Poiseuille from the inlet
- 2 **What the gas carries** — the species equation, its solution
- 3 **Every speed** — one PINN, and the speed a design needs

White carries the gas, Incropera the mass transfer and Liu the method; the example is Ex_09.2's fuel channel; L10.2 adds the electrode behind it.

READ, AND WHAT EACH ANSWERS

White ch. 9 when is a gas incompressible?

Incropera ch. 14 how is a species carried?

Liu ch. 2.2 what does an input speed buy?

The Fuel Channel

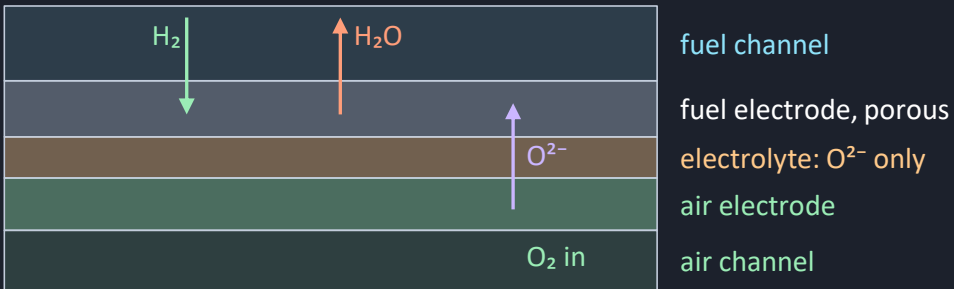
Fuel channel — the gap between a solid oxide cell's fuel electrode and its interconnect. Oxide ions crossing from the air side burn the hydrogen in the porous electrode, and the steam made returns to the gas: one molecule each way for every two electrons.

THE REACTION, AND THE FLUX IT SETS

$$N = \frac{i}{2F}$$
$$\text{H}_2 + \text{O}^{2-} \rightarrow \text{H}_2\text{O} + 2\text{e}^-$$

N molecules exchanged per area and second, i the current density, F Faraday's constant

BEHIND THE WALL: THE FUEL CELL



THE CHANNEL OF EX_09.2

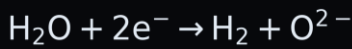
	value
channel	100 mm long, 1 mm high
gas in	97 % H ₂ , 3 % H ₂ O
state	800 °C, 1 atm: 11.36 mol/m ³
current	0.5 A/cm ² : $N = 0.0259 \text{ mol}/(\text{m}^2 \text{ s})$
inlet speed	0.3 to 1.5 m/s

Seen from the gas only; L10.2 adds the electrode. The diffusivity of steam in hydrogen, $8.0\text{e-}04 \text{ m}^2/\text{s}$, is an estimate and the current a typical value.

Electrolysis Mode

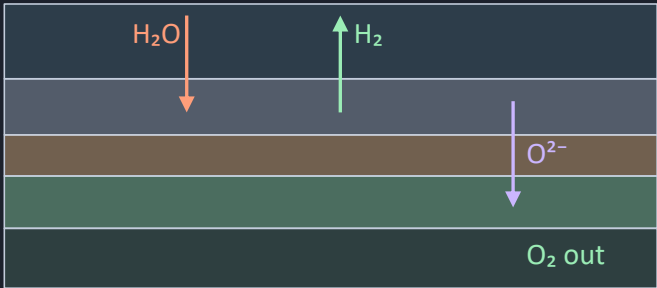
Solid oxide electrolysis — the same cell driven backwards by a power supply. Steam is split at the fuel electrode, the oxide ions cross the electrolyte to the air side and leave as oxygen, and hydrogen returns to the gas. Steam is now the reactant, so the feed is mostly steam.

THE REACTION, RUN BACKWARDS



O^{2-} an oxide ion, through the electrolyte, e^- an electron, round the circuit

BEHIND THE WALL: THE ELECTROLYSER



fuel channel
fuel electrode, porous
electrolyte: O^{2-} only
air electrode
air channel

FUEL CELL AGAINST ELECTROLYSER

	fuel cell	electrolyser
fed	97 % H_2 , 3 % H_2O	90 % H_2O , 10 % H_2
taken	H_2	H_2O
returned	H_2O	H_2
the minor gas	fixes the voltage	keeps nickel reduced
power	out, to a load	in, from a supply

Steam is fed to a fuel cell not because the reaction needs it - the reaction makes it - but because the voltage depends on the ratio of hydrogen to steam, infinite with none. Typical feeds, not cited. For the channel only the flux changes sign.

Incompressible Gas

Incompressible gas — a gas whose speed is far below the speed of sound, so that the pressure differences that drive it change its density by a negligible amount. The measure is the Mach number Ma ; below about 0.3 the flow may be treated as incompressible, like a liquid.

THE MACH NUMBER, AND AN IDEAL GAS

$$Ma = \frac{u}{a}, \quad a = \sqrt{\frac{\gamma RT}{M}}, \quad \frac{\Delta \rho}{\rho} \approx \frac{Ma^2}{2}$$

$$c = \frac{p}{RT}$$

a the speed of sound, γ the ratio of heat capacities, M the molar mass
 c moles per volume, p pressure, R the gas constant, T temperature

SPEED, LOG SCALE



THE FUEL GAS AT 800 °C

quantity	value
molar mass M	2.5 g/mol
speed of sound a	about 2240 m/s
Mach number	below 10 ⁻³
density change	about 10 ⁻⁷
moles per volume c	11.36 mol/m ³

Light hydrogen at 800 °C carries sound fast. The flow in the channel is as incompressible as water in a tube; $\gamma = 1.4$ is taken for the diatomic gas.

Equimolar Exchange

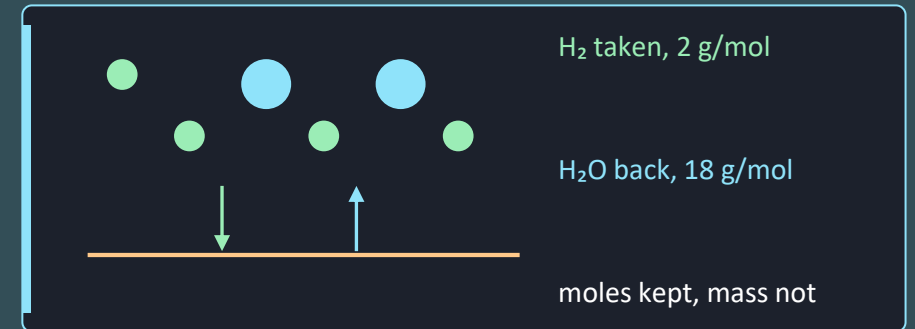
Equimolar exchange — a reaction at the wall that removes as many molecules from the gas as it returns. In the fuel channel one hydrogen leaves for every steam that enters, so the number of moles per volume stays p/RT and the velocity is not changed by the reaction.

COUNT IN MOLES, NOT IN MASS

- **moles** — one out, one in: unchanged
- **mass** — light H_2 out, heavy H_2O in: rising
- **so** — the species is counted in mole fractions

TALK · 2 MIN How would you explain, in your own words, why the reaction does not push the gas faster? Then, in the drawing on the right, what would change if two molecules entered for each one that left?

ONE MOLECULE FOR ONE



The density of the gas rises along the channel as hydrogen becomes steam, while its molar concentration and its velocity stay the same. A mass fraction would mix the two effects; a mole fraction separates them.

Poiseuille From the Inlet

Developed channel flow — the flow between two plates far from the inlet: the parabola of L9.1 in a flat channel. Here the Reynolds number is 1.3 and the entrance length 0.06 mm, so the gas is fully developed from the inlet and its velocity is known everywhere before anything is solved.

THE PARABOLA, AND ITS PRESSURE DROP

$$u(z) = 6\bar{u} \frac{z}{h} \left(1 - \frac{z}{h}\right), \quad \Delta p = \frac{12\mu\bar{u}L}{h^2}$$

z across the channel, h its height, L its length, $\bar{u}$ the mean speed, μ viscosity

ACROSS THE CHANNEL



zero on both walls

1.5 times the mean in the middle

formed within 0.06 mm

THREE FACTS, THREE NUMBERS

fact	because
incompressible	Mach below 10^{-3}
velocity unchanged	one molecule for one
parabola from the inlet	Re 1.3, entrance 0.06 mm

Poiseuille's pressure drop over the channel is 26 Pa at 1 m/s. Darcy's law of L9.1 with $K = h^2/12$: the flow is solved, and the rest of the lecture is what it carries.

Convection–Diffusion

Convection–diffusion — the equation for a species carried by a known flow: the flow moves it along, diffusion spreads it from rich to poor. Its conditions are the inlet's composition, the flux the electrode returns, and an impermeable interconnect.

THE STEAM FRACTION, AND ITS THREE CONDITIONS

$$u(z) y_x = D y_{zz}$$

$$y = y_{in} \text{ at } x = 0, \quad -Dc y_z = N \text{ at } z = 0, \quad y_z = 0 \text{ at } z = h$$

y the steam fraction, x along, z across, D the diffusivity, N the wall flux

THE TWO WAYS A SPECIES MOVES



ALONG AGAINST ACROSS

$$Pe_L = \frac{\bar{u}L}{D} = 125$$

along carried, Péclet 125

across spread, Péclet 1.25

dropped diffusion along: 10^{-4}

The Péclet number compares carrying with spreading. Along the channel the flow wins by a hundred and twenty-five; across its height diffusion keeps up.

Marching Along the Channel

Parabolic equation — an equation with a direction in which information travels one way only. Without diffusion along it, the species equation is parabolic in x : it marches from the inlet as the heat equation of L8.2 marches in time, and it needs no condition at the outlet.

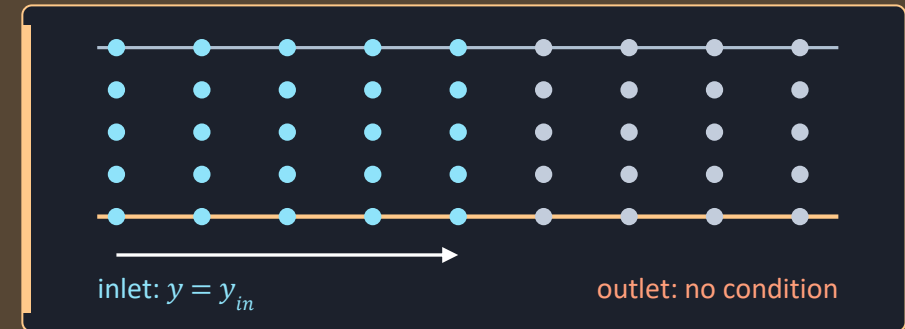
THE SCALED EQUATION, MARCHED

$$6Z(1-Z)Gs\theta_x = \theta_{zz}, \quad G = \frac{u_{ref}h^2}{DL}$$

$X = x/L$, $Z = z/h$, $s = \bar{u}/u_{ref}$, θ the rise of y in units of the channel's rise at u_{ref}

TALK · 2 MIN How would you decide, in your own words, which conditions an equation needs and where? Then, in the drawing on the right, why would a zero-slope condition at the outlet be wrong for this channel?

ONE COLUMN AT A TIME



FDM: Crank–Nicolson along, two implicit steps first, central differences across; 401×41 nodes meet 0.1 percentage point at every speed in a few ms. The mean keeps rising to the last millimetre, so a zero-slope outlet would be wrong.

Fully Developed Concentration

Fully developed concentration — the state beyond a short entrance in which the profile across the channel keeps its shape while the whole of it rises along. The rise of the mean is a balance: what the wall returns, the flow carries away.

THE BALANCE, AND THE EXACT FIELD

$$\bar{y}(x) = y_{in} + \frac{Nx}{\bar{u}hc}$$

$$\theta = \frac{X}{s} + G \varphi(Z), \quad \varphi = Z^3 - \frac{Z^4}{2} - Z + \frac{13}{35}$$

$\bar{y}$ the mixing cup mean, X/s that balance, φ the profile across, zero mean

ACROSS THE CHANNEL, AT ANY x

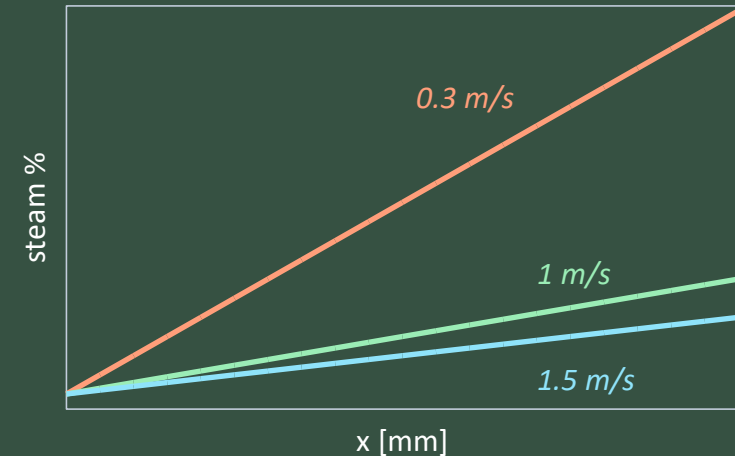


interconnect: a little less

dashed: the mean

electrode: $13G/35$ above it

STEAM ALONG THE CHANNEL



Ex_09.2's exact field: from 3 % to 26 % at 1 m/s, to 79 % at 0.3 m/s. The electrode sees 0.11 percentage points more than the mean.

The Sherwood Number

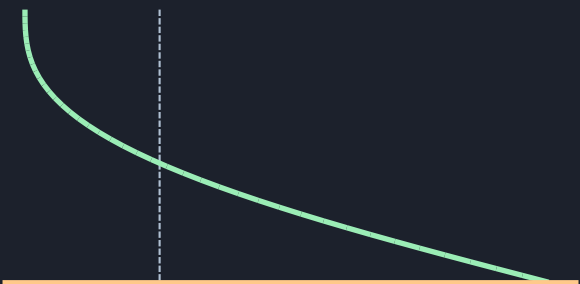
Sherwood number — the mass-transfer coefficient between a wall and the mean of the gas, made dimensionless with the channel's size and the diffusivity. It is the Nusselt number of heat transfer for a species: in a fully developed channel it is a constant of the geometry.

THE WALL AGAINST THE MEAN

$$Sh = \frac{h_m 2h}{D} = \frac{2 \cdot 35}{13} = 5.385$$

h_m the mass transfer coefficient, $2h$ the hydraulic diameter, D the diffusivity

THE GAP IT MEASURES



interconnect: a little less

dashed: the mean

electrode: 13G/35 above it

WHAT EACH METHOD FINDS

	Sherwood
exact	5.385
FDM, 401 × 41	3.71 to 4.94
PINN	6.12 to 6.75
heat, same plates	Nusselt 5.385

One wall at a constant flux, the other impermeable: Shah & London's constant, and Incropera's table 8.1 for heat. Both methods find the field to a ten-thousandth but the Sherwood number only to 8 to 31 %: it is a tenth of a per cent of the field.

One PINN for Every Flow Rate

Parametric PINN — a PINN that takes a parameter of the problem as an input beside the coordinates, here the inlet speed s . It is trained once over a range of speeds, and after that any speed is read off, not solved again: the one thing a march cannot do.

THE BALANCE BUILT IN, THE PROFILE LEARNT

$$\hat{\theta}(X,Z,s) = \frac{X}{s} + G \mathcal{N}(X,Z,s)$$

X/s the balance, built in, $\mathcal{N}$ the network: the profile in units of G

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=3, n_hidden=32, n_layers=4)      # X, Z and the speed s in
th = X / s + G * net(torch.cat([X, Z, s], 1))  # the balance, built in
thX = grad(th, X); thZ = grad(th, Z)           # derivatives by autograd
r = (grad(thZ, Z) - 6*Z*(1 - Z)*G*s*thX) / G  # residual, of order one
```

FDM AND PINN, SCORED EXACTLY

	FDM	PINN
as exact	401 × 41 nodes	1.7e-04
the field	3.4 ms	1.5 ms
one number	3.4 ms	about 0.1 ms
new speed	a new march	read off
offline	none	2 minutes

Online, inference only, at the same accuracy (1 m/s), on a laptop CPU: two times faster for the field, 30 to 40 for one number. The training is repaid only after tens of thousands of answers; the speed as an input is the larger gain.

Design by Gradient

Design by gradient — finding the parameter a design rule asks for from a PINN that has it as an input: the PINN gives the quantity, automatic differentiation its derivative with respect to the parameter, and Newton's method the value, with no further solve.

NEWTON THROUGH THE PINN

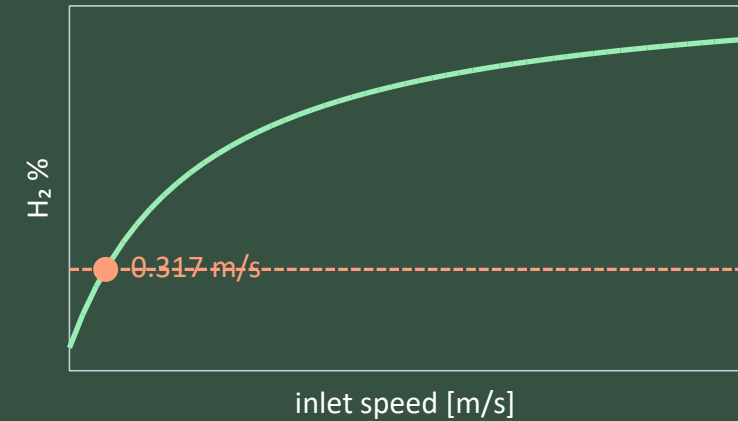
$$s_{k+1} = s_k - \frac{f(s_k) - f^*}{f'(s_k)}$$

$f(s)$ the hydrogen at the electrode at the outlet, $f^* = 25\%$, f' by autograd

THE DESIGN SPEED IN PYTORCH

```
s = (1 / theta_hand).requires_grad_()      # start: the hand calculation
for k in range(5):
    f = h2_at_electrode_outlet(net, s)      # the network, at X = 1, Z = 0
    s = s - (f - 0.25) / grad(f, s)         # Newton, the slope by autograd
```

25 % HYDROGEN AT THE OUTLET



PINN	0.3173 m/s, five steps
exact	0.3174 m/s
then	74 % used, 8 Pa

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
turbulence	eddies of every size	Pope ch. 6	every fast flow
turbulence models	the mean, and a model for the rest	Pope ch. 4, 10	industrial CFD
<u>Navier–Stokes problem</u>	can a smooth flow break down?	Pope ch. 6	the mathematics
the Graetz entrance	the first millimetre of the profile	Incropera ch. 8	short channels
several species	each diffusing through the others	Incropera ch. 14	the air side, MP9.2B
compressible gas	density with the pressure	White ch. 9	Mach above 0.3

WHY THEY ARE LEFT OUT

The fuel channel's gas is slow, laminar and fully developed, with an exact solution to score against, so none of them is needed to do Ex_09.2. The first three rows are turbulence, which no flow in this cell reaches.

Exercise 9.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the fuel channel, its data, the exact field
- **4** — finite differences, marched from the inlet
- **5** — one PINN for every speed; timed at equal accuracy
- **6** — the inlet speed a design asks for
- **7 – 9** — the report; mini projects MP9.2A and MP9.2B

THE QUESTION EACH PART ANSWERS

- 1 – 3** why is the gas simple?
- 4** how fine must the march be?
- 5** what does a PINN add?
- 6** which speed keeps 25 %?

REPORT QUESTION, TO CONCLUDE

At the same accuracy the PINN answered two to forty times faster online, after two minutes of training, and found the design speed without a sweep. For which questions about this cell would you train a PINN, and for which march?

Key Takeaways

GAS AND SPECIES

what it is	a slow gas carrying a species
function	the steam fraction everywhere
train, tune	exact: balance plus a quartic
vs a liquid	the same flow; moles, not mass
used for	fuel use, the electrode's gas

ONE PINN, EVERY SPEED

what it is	a PINN with the speed as input
function	the field at any speed, read off
train, tune	the balance built in; once
vs a march	faster online, no re-solve
used for	design by gradient

Next — L10.1: the species equation again, inside a battery's particle.

Questions

1 – 2 · GAS AND SPECIES

1 Which three facts make the gas in a fuel channel a known, incompressible flow?

Its Mach number is below a thousandth, so its density hardly changes. The electrode returns one molecule for each it takes, so the moles per volume and the velocity stay the same. And the Reynolds number is 1.3, so Poiseuille's parabola holds from the inlet. *White ch. 9; Ex_09.2 section 3*

2 Why can diffusion along the channel be dropped, and what does that make the equation?

The Péclet number along it is 125, so that term is a ten-thousandth of the one across. Without it the equation is parabolic in x: it marches from the inlet like heat in time and needs no outlet condition. A zero-slope outlet would be wrong, as the mean keeps rising. *Incropera ch. 14; Ex_09.2 sections 3 – 4*

3 – 4 · ONE PINN, EVERY SPEED

3 What does the PINN's trial function build in, and why is the Sherwood number hard to get?

The balance X/s , the hand calculation, so the network learns only the profile across, in units of G . Both methods find the field to a ten-thousandth, but the Sherwood number only to within 8 to 31 %, because the profile it measures is a tenth of a per cent of the field. *Incropera ch. 8; Ex_09.2 section 5*

4 What can a PINN with the speed as an input do that a march cannot?

Give the field at any speed without solving again, and its derivative with respect to the speed by autograd. Newton's method then finds the speed for 25 % hydrogen at the outlet, 0.3173 m/s against the exact 0.3174, with no sweep. At equal accuracy it also answers two to forty times faster. *Liu ch. 2.2; Ex_09.2 section 6*

NEXT

L10.1 — Ionic Diffusion

The species equation again, inside the particle of a battery's electrode, where the measurement is a voltage at its surface.

BEFORE THE NEXT SESSION

read	Incropera ch. 14
run	Ex_09.2, sections 1 to 7
bring	your design speed, and its error
think about	a parameter you would make an input

Block nine is finished: a liquid in a tube, and a gas that carries something along a channel.